

Shivani Kamtikar

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Ph.D. Candidate, Computer Science · University of Illinois Urbana-Champaign

RESEARCH SUMMARY

I develop learning-based methods for manipulation with soft, rigid and hybrid robotic arms, with an emphasis on dexterous control in cluttered and unstructured environments. I also work on tactile-centric manipulation policies using anthropomorphic hands. Recently, I've been working on world models to predict robot dynamics from vision for various downstream manipulation tasks. My research has broad applications, including autonomous harvesting, human-assistive devices, medical imaging, underwater exploration, manufacturing, and more, with publications at RA-L, IROS, ICRA, and RoboSoft.

EDUCATION

University of Illinois at Urbana-Champaign

May 2022 – Dec 2026 (expected)

Ph.D. in Computer Science · Advisor: Prof. Girish Chowdhary

Relevant Courses: Robot Learning, 3D Vision, Transfer Learning, Meta Learning

University of Illinois at Urbana-Champaign

May 2022

M.S. in Computer Science · Advisor: Prof. Girish Chowdhary

Relevant Courses: Machine Learning, Computer Vision, Learning-Based Robotics, Autonomous Systems

Savitribai Phule Pune University

August 2020

B.Tech. in Information Technology

PUBLICATIONS

Conference / Journal (in order of relevance)

- **S. K. Kamtikar**, P. Asthana, N. K. Uppalapati, G. Krishnan, G. Chowdhary. "THREAD: Trajectory Planning for Hybrid Rigid-Soft Manipulators with Environment-Aware Diffusion." **IROS 2026 [Oral Presentation]**
- **S. K. Kamtikar**, C. H. Kim, J. Migdal, C. Tabasso, T. Brady, T. Padir. "TACTFUL: Tactile-Driven Exploration for Object Localization and Identification in Confined Environments." **IROS 2026 [Oral Presentation]**
- **S. K. Kamtikar**, K. Koe, J. Wasserman, S. Marri, B. T. Walt, N. K. Uppalapati, G. Krishnan, G. Chowdhary. "HyReach: Vision-Guided Hybrid Manipulator Reaching in Unseen Cluttered Environments." **Soft Robotics Journal 2026**
- **S. K. Kamtikar**, S. Marri, B. T. Walt, N. K. Uppalapati, G. Krishnan, G. Chowdhary. "Visual Servoing for Pose Control of Soft Continuum Arm in a Structured Environment." **IEEE Robotics and Automation Letters (RA-L) + RoboSoft 2022 [Oral Presentation]**
- C. H. Kim, **S. K. Kamtikar**, T. Brady, T. Padir, J. Migdal. "Grasp, Slide, Roll: Comparative Analysis of Contact Modes for Tactile-Based Shape Reconstruction." **ICRA 2026 [Poster Presentation]**
- K. Koe, S. Marri, B. Walt, **S. K. Kamtikar**, N. K. Uppalapati, G. Krishnan, G. Chowdhary. "Learning-Based Position and Orientation Control of a Hybrid Rigid-Soft Arm Manipulator." **ASME Journal of Mechanisms and Robotics 2025**
- K. Koe, P. K. Shah, B. Walt, J. Westphal, S. Marri, **S. K. Kamtikar**, J. S. Nam, N. K. Uppalapati, G. Krishnan, G. Chowdhary, "Precision Harvesting in Cluttered Environments: Integrating End Effector Design with Dual Camera Perception." **ICRA 2025**

Workshop Papers

- **S. K. Kamtikar** et al. "Visual Servoing for Pose Control of Hybrid Continuum Manipulator in an Unstructured Environment." **CoRL 2023 Workshop on Learning for Soft Robots**.
- **S. K. Kamtikar** et al. "Realistic Simulation Environments to Achieve Visual Servoing on Soft Continuum Arms in Constrained Environments." **MLCAS 2022**.
- **S. K. Kamtikar** et al. "Towards Autonomous Berry Harvesting using Visual Servoing of Soft Continuum Arm." **AI4FS Workshop 2022**.

WORK EXPERIENCE

Applied Scientist II Co-op - Amazon Robotics

May 2026 – Present

Mentor: Dr. Harpreet Sawhney and Dr. Mudit Agrawal

- Developed a conditional flow-matching world model with JEPA-style latent supervision to predict manipulation dynamics for various downstream manipulation tasks (paper in preparation)

Applied Scientist II Co-op - Amazon Robotics

Jan 2025 – Jul 2025

Mentor: Prof. Taskin Padiir and Dr. Josh Migdal

- Developed a vision-free autonomous tactile exploration framework enabling anthropomorphic hands (Inspire Dexterous FTP) to localize and identify objects in confined environments (accepted at IROS 2026)
- Designed a learned policy combining behavior cloning and reinforcement learning (RL) using tactile point clouds for 3D object reconstruction
- Introduced a dynamic reward schedule balancing global exploration and local interaction; achieved lower Chamfer distances and higher success rates vs. baselines
- Advanced tactile sensing as a primary perception modality for robotic manipulation in unstructured settings
- Conducted a study on how different contact modes affect object shape reconstruction using a tactile-enabled dexterous gripper (accepted at ICRA 2026)

RESEARCH EXPERIENCE

Diffusion-Based Path Planning for Hybrid Continuum Manipulators

- Designed the first ever diffusion-based generative planner for feasible backbone trajectories of hybrid soft-rigid manipulators, conditioned on scene geometry and goal constraints (accepted at IROS)
- Developed a unified shape-aware motion planning framework producing low-strain, collision-free trajectories in confined spaces
- Implemented inference-time collision avoidance with online reinforcement learning (RL) refinement
- Demonstrated sample-efficient cross-embodied real-world transfer in hybrid robots

Contact-Driven Planning via Vision-Language Models (VLMs)

- Developed a VLM-guided, contact-aware planning framework for hybrid manipulators, enabling intentional bracing and wrapping around obstacles
- Designed a diffusion-based motion generator conditioned on VLM-predicted contact affordances for confined-space manipulation
- Demonstrated extended workspace reach, reduced actuator strain, and improved stability near obstacles

3D Vision-Guided Autonomous Manipulation of Hybrid Robots

- Developed a real-time open-world reaching system for rigid-soft continuum manipulators in unstructured environments (accepted at Soft Robotics Journal)
- Enabled obstacle avoidance through 3D reconstruction and shape-informed path planning without expensive sensors
- Achieved 98% pose accuracy with a novel learned controller for the hybrid arm system

Learning-Based Control of Soft Robotic Arms

- Built a deep neural network for robust 3D positioning of soft robotic arms using visual feedback from a distal-end camera (accepted at RA-L & RoboSoft)
- Achieved state-of-the-art performance: translation error < 2 cm and rotation error < 0.25 rad
- Demonstrated transferability to new environments with minimal retraining, enabling scalable deployment

TECHNICAL SKILLS

Research Areas: Robotics (World Models, Tactile Sensing, RL, Imitation Learning, Visual Servoing, Soft Robotics, Manipulation), 3D Computer Vision (SLAM, 3D Reconstruction, RGB-D Fusion), Generative Models (Diffusion Models, VLMs, Transformers) for Robot Planning

Programming & Tools: Python, PyTorch, ROS, ROS2

AWARDS & RECOGNITION

- Future Leaders of Robotics and AI Award - University of Maryland, College Park, 2025
- Student Leadership Award (GIANT Award) - IDEA Institute, UIUC, 2025
- Patent granted by the Indian Patent Office for Indian Sign Language Recognition system, 2024
- IBM Research Grant for Best Research Project - Undergraduate final-year project, 2021
- Best Outgoing Student Award - Savitribai Phule Pune University

INVITED TALKS

- Future Leaders of Robotics and AI Seminar Series - University of Maryland, College Park, 2025
- MLCAS Fourth International Workshop on ML for Cyber-Agricultural Systems - 2022
- AI for Agriculture and Food Systems (AIAFS) Workshop - 2022

SERVICE & LEADERSHIP

Teaching Assistant

- CS 102: Little Bits to Big Ideas - UIUC (Jan 2026 - May 2026)
- CS 498: Introduction to Deep Learning - UIUC, with Prof. Lana Lazebnik (Sep-Dec 2020)

Reviewing:

- IEEE Robotics and Automation Letters (RA-L)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- IEEE India Council International Conference (INDICON) 2024

Community:

- GradSWE Director, UIUC - 2025–2026
- weSTEM 2025 Director, UIUC
- Computer Science Representative, Engineering Graduate Student Advisory Council (EGSAC) – 2023-2024
- General Chair, CSL Student Conference 2024
- Robotics Chair, CSL Student Conference 2023